

A Rigorous Treatment of the Beamsplitter and the Total Matrix of a Mach–Zehnder Interferometer (MZI)

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1 Introduction

A beamsplitter (BS) is an essential linear optical component, used to split and combine electromagnetic waves in both classical and quantum optical setups. A *Mach–Zehnder Interferometer* (MZI) consists of two such beamsplitters plus two arms, at least one of which can introduce a controlled phase shift (e.g. via free-space path difference or an active modulator). In many practical MZIs, reflections from mirrors in one (or both) arms also contribute fixed or additional phase shifts.

In this document:

- We review the beamsplitter as a linear, *unitary* 2×2 transformation.
- We discuss its 50:50 (balanced) case, including a typical matrix representation with specific phase factors for reflection and transmission.
- We derive the *overall* matrix of an MZI, including phase shifts from the arms and/or mirrors.

2 Beamsplitter as a Linear Transformation

2.1 Fields and Amplitudes

Let E_1 and E_2 be the complex field amplitudes entering the two input ports of a beamsplitter. We may arrange them in a column vector

$$\begin{pmatrix} E_1 \\ E_2 \end{pmatrix}. \tag{1}$$

Similarly, denote the two output port amplitudes by E'_1 and E'_2 , forming

$$\begin{pmatrix} E'_1 \\ E'_2 \end{pmatrix}. \quad (2)$$

A general *linear, lossless* beamsplitter transformation can be written as

$$\begin{pmatrix} E'_1 \\ E'_2 \end{pmatrix} = U_{\text{BS}} \begin{pmatrix} E_1 \\ E_2 \end{pmatrix}, \quad (3)$$

where U_{BS} is a 2×2 *unitary* matrix.

2.2 Unitarity from Losslessness

A *lossless* beamsplitter preserves total optical power (or intensity). Classically, we require

$$|E_1|^2 + |E_2|^2 = |E'_1|^2 + |E'_2|^2 \quad \text{for all } (E_1, E_2). \quad (4)$$

By substituting Eq. (??), one sees that this equality for *all* input amplitudes is exactly the statement

$$U_{\text{BS}}^\dagger U_{\text{BS}} = I_{2 \times 2},$$

meaning U_{BS} is **unitary**. In a quantum framework (where each E_j might be replaced by annihilation operators $\hat{a}_j$), the same requirement enforces that photon number (and commutation relations) be preserved, again leading to unitarity.

2.3 50:50 (Balanced) Splitting

A *balanced* beamsplitter sends half the power to each output port if *only one* port is excited at the input. Concretely:

$$|E'_1|^2 = |E'_2|^2 = \frac{1}{2} |E_{\text{in}}|^2 \quad \text{when } \begin{pmatrix} E_1 \\ E_2 \end{pmatrix} = \begin{pmatrix} E_{\text{in}} \\ 0 \end{pmatrix}.$$

Likewise if $E_1 = 0$ and $E_2 \neq 0$. That enforces $|r|^2 = |t|^2 = \frac{1}{2}$ where r, t are the reflection and transmission amplitudes, respectively.

2.4 Example: 50:50 Matrix with a $\pi/2$ Phase Shift

A commonly encountered balanced beamsplitter matrix is

$$U_{\text{BS}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}, \quad (5)$$

which satisfies:

$$U_{\text{BS}}^\dagger U_{\text{BS}} = I, \quad |r| = |t| = \frac{1}{\sqrt{2}}, \quad r = \frac{i}{\sqrt{2}}, \quad t = \frac{1}{\sqrt{2}}.$$

The i (imaginary unit) factor indicates a relative $\pi/2$ phase shift in one of the reflection paths, a typical outcome of partial reflection from a thin-film coating. Different physical devices can yield different phase conventions.

3 Mach–Zehnder Interferometer (MZI)

3.1 Components

- **Two beamsplitters:** BS_1 and BS_2 , each described by a (possibly identical) unitary 2×2 matrix, often 50:50.
- **Two arms (paths):** The outputs of BS_1 travel along two distinct paths before recombining at BS_2 .
- **Phase shifts:**
 1. *Propagation phase* due to path length difference (or any additional phase modulator).
 2. *Mirror reflection(s)* in one or both arms, possibly introducing a fixed phase of π (or -1 factor) per reflection.

3.2 Matrix Picture of the MZI

First Beamsplitter. Call it U_{BS_1} . It acts on the input amplitudes $\begin{pmatrix} E_1 \\ E_2 \end{pmatrix}$ to yield the two fields emerging from BS_1 .

Path Phases and Mirror Reflection. Suppose the top path acquires a total phase ϕ_{top} , while the bottom path has phase ϕ_{bot} . In matrix form, we can represent these phases via a *diagonal* matrix

$$D(\phi_{\text{top}}, \phi_{\text{bot}}) = \begin{pmatrix} e^{i\phi_{\text{top}}} & 0 \\ 0 & e^{i\phi_{\text{bot}}} \end{pmatrix}. \quad (6)$$

For example, a mirror reflection in the top arm might add a fixed π phase ($e^{i\pi} = -1$) or some design-specific phase shift. Meanwhile, if the bottom arm has a different path length, that contributes $e^{i\phi_{\text{bot}}}$. One often normalizes one path to 0 phase, letting the other path carry *relative* phase $\phi = \phi_{\text{top}} - \phi_{\text{bot}}$.

Second Beamsplitter. Call it U_{BS_2} . It recombines the two arms into the final output ports.

Hence, the full MZI transformation from inputs $\begin{pmatrix} E_1 \\ E_2 \end{pmatrix}$ to outputs $\begin{pmatrix} E'_1 \\ E'_2 \end{pmatrix}$ is

$$\begin{pmatrix} E'_1 \\ E'_2 \end{pmatrix} = U_{BS_2} D(\phi_{\text{top}}, \phi_{\text{bot}}) U_{BS_1} \begin{pmatrix} E_1 \\ E_2 \end{pmatrix}. \quad (7)$$

If U_{BS_1} and U_{BS_2} are each unitary, and $D(\phi_{\text{top}}, \phi_{\text{bot}})$ is also unitary (diagonal phases), then their product

$$U_{\text{MZI}} \equiv U_{BS_2} D(\phi_{\text{top}}, \phi_{\text{bot}}) U_{BS_1}$$

is itself a unitary 2×2 matrix.

3.3 Including Mirror-Induced Phases Explicitly

Schematic of an MZI with Mirrors:
 One or both arms may reflect off a mirror, incurring a phase factor -1 (i.e. $e^{i\pi}$) each time. Additional path-length or modulator-induced phase shifts can appear as well. The arms recombine on BS_2 .

Figure 1: Mach–Zehnder Interferometer with possible mirror reflections adding fixed phases.

Often, the top path has a mirror that adds a phase factor $e^{i\pi} = -1$, and the bottom path has either no mirror or a different reflection count. We can incorporate all those reflection phases into ϕ_{top} and ϕ_{bot} . For example:

$$\phi_{\text{top}} = \phi_{\text{path}} + \pi, \quad \phi_{\text{bot}} = \phi_0,$$

where ϕ_{path} is a tunable phase difference from changing path length, and π arises from reflection off the mirror in the top arm. One can thereby account for how each reflection or transmission event might contribute an i factor (for a partial reflection on the beamsplitter) or -1 factor (for a mirror reflection), etc.

In short, *any* stable optical phase shift may be collected into $D(\phi_{\text{top}}, \phi_{\text{bot}})$, and the overall MZI matrix remains

$$U_{\text{MZI}} = U_{\text{BS}_2} D(\phi_{\text{top}}, \phi_{\text{bot}}) U_{\text{BS}_1}.$$

4 Explicit Example of the Overall MZI Matrix

For a concrete example, choose the same 50:50 beamsplitter matrix for both BS_1 and BS_2 as in Eq. (??), i.e.:

$$U_{\text{BS}_1} = U_{\text{BS}_2} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}.$$

Let the *top* arm gain a total phase ϕ_t and the *bottom* arm gain ϕ_b . Then

$$D(\phi_t, \phi_b) = \begin{pmatrix} e^{i\phi_t} & 0 \\ 0 & e^{i\phi_b} \end{pmatrix}.$$

Thus,

$$U_{\text{MZI}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \begin{pmatrix} e^{i\phi_t} & 0 \\ 0 & e^{i\phi_b} \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}.$$

One can multiply these matrices to obtain the explicit 2×2 form of U_{MZI} , from which one may deduce how (E_1, E_2) map to (E'_1, E'_2) for arbitrary phases ϕ_t and ϕ_b . In many treatments, one sets $\phi_b = 0$ for simplicity and looks at the *relative* phase $\phi = \phi_t - \phi_b$. Perfect constructive or destructive interference in one output port is then observed as ϕ is varied.

5 Conclusion

- **Beamsplitter:** A lossless 2×2 *unitary* matrix that linearly transforms two input fields to two output fields. A 50:50 beamsplitter has $|r|^2 = |t|^2 = \frac{1}{2}$, e.g. Eq. (??).
- **Mach–Zehnder Interferometer:** Formed by two beamsplitters and two arms. Each arm can include phase shifts from mirrors and/or path length differences, modeled by a diagonal unitary matrix $D(\phi_t, \phi_b)$.
- **Overall Matrix:** The MZI transformation is

$$U_{\text{MZI}} = U_{\text{BS}_2} D(\phi_t, \phi_b) U_{\text{BS}_1},$$

which remains unitary (energy-conserving). Detailed interference patterns at the outputs arise from the interplay of beamsplitter reflections/transmissions and the phase factors $e^{i\phi_t}$, $e^{i\phi_b}$, including mirror-induced π shifts.

This linear-algebraic approach explains quantitatively the interference fringes observed in an MZI, how mirror reflections contribute fixed phases, and how controllable path differences shift the output intensities between the two final ports. In both classical and quantum optics, such an analysis underlies many fundamental interference experiments.

References

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